Scam Guardian

A calm guardian that listens to the scam call **while it is happening** and tells a senior to **hang up**, in their own language.

The problem

When the children are at work, the “official” calls their parents instead. Callers pose as **police, CBI or bank staff**, claim a “digital arrest”, and use **fear, urgency and secrecy** to get an **OTP** or a payment. Over **Rs 3,000 crore** has reportedly been lost in India to “digital arrest”, which the I4C advisory of 6 March 2025 states **does not exist in Indian law**. Number-blocking and call-labelling tools do not hear what is said, and fraud reports come after the money has gone. Nothing helps a parent **in the middle of the call**, in **Hindi, Tamil or Malayalam**.

Our solution

An old phone or tablet sits beside the senior’s phone during a call on **speakerphone**. Scam Guardian hears the conversation, finds scam tactics, and shows a large three-level signal: **No warning detected**, **Be careful** or **Hang up now**, with a spoken warning in the senior’s language. Family receive the **reasons only**, never audio or quotes.

How it works

- **Hears the call:** the browser streams audio to **Gnani Prisma v3** realtime speech recognition in code-mix mode, so **Hinglish, Tanglish and Manglish** work too.
- **Knows who is speaking:** a one-time **30-second voice enrollment** and a phone-line frequency test label each turn **You**, **Caller** or **Unsure**. A voice that sounds like the senior but arrives through the phone line is **never trusted**. Only a small numeric measurement is kept; the audio is discarded.
- **Spots tactics:** language-pack rules (authority, threat, urgency, secrecy, payment, OTP, remote-access apps) **plus a compulsory AI check**. Offers alone do not raise an alarm; only a request for money or secrets does, to keep false alarms down.
- **Warns and tells family:** full-screen, high-contrast alert not reliant on colour; spoken warning through **Gnani Timbre**; optional family alerts with a code word and a trusted-voice message.

Why it is different

- **During the call**, not after the loss.
- **Four languages** and code-mixed speech: English, Hindi, Tamil, Malayalam.
- **Senior-first design:** large type, one main button, no scores or jargon.
- **Privacy by design:** no audio stored, transcripts only in memory for the session, family see **reasons, not conversations**.

Honest status and next steps

A **working MVP** with 138 automated tests, built on synthetic data. It is **not yet validated on real calls**. Next: **pilot with older adults**, measure speech accuracy on real speakerphone audio, calibrate speaker labels, native-speaker review of Hindi, Tamil and Malayalam wording, and **legal advice** on transcribing the remote caller. Guardian cannot end a call or block payments; it tells the senior when to.

Code, demo video and run guide: github.com/johhhnnnnyyyyy/Scam-Guardian